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NIHR Research for Patient Benefit Programme

Scheme	Research for Patient Benefit Programme
Funding Opportunity	2025/334 RfPB Researcher-led

SUMMARY INFORMATION

Application Title	Trust-wide behavioural science feasibility study on catheterisation duration and infection reduction
Contracting Organisation	KING'S COLLEGE HOSPITAL NHS FOUNDATION TRUST
Contracting Organisation Department	King's College Hospital NHS Foundation Trust
Duration (months)	18
Total Cost to NIHR	£233,491.37
Has this application been previously submitted to this or any other funding body?	No

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Mrs Ashley Flores
Organisation	KING'S COLLEGE HOSPITAL NHS FOUNDATION TRUST
Department	Infection Prevention & Control
Proposed Role	Responsible for the overall leadership and practical management of the study, ensuring regulatory and ethical approvals are in place, and will be main point of contact. They will act as coordinator and will ensure the project is conducted according to the approved plan and reporting requirements.
ORCID	0000-0002-4901-8207
% FTE Commitment	100.00%

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Joint Lead Applicant

Full Name	Ms Rebecca Rowe
Organisation	Revealing Reality LtdRevealing Reality Ltd
Department	ResearchResearch
Proposed Role	Becky will manage day-to-day project delivery. Collaborating with the research team, she will apply behavioural science & healthcare design expertise. She will oversee project elements, including co-design, data collection, analysis, and quality assurance, and manage the RR team.Becky will manage day-to-day project delivery. Collaborating with the research team, she will apply behavioural science & healthcare design expertise. She will oversee project elements, including co-design, data collection, analysis, and quality assurance, and manage the RR team.
ORCID	0009-0008-9523-27310009-0008-9523-2731
% FTE Commitment	45.00%45.00%

Co-Applicants

Full Name	Dr Martin Brown
Organisation	KING'S COLLEGE HOSPITAL NHS FOUNDATION TRUST
Department	Medical Microbiology
Proposed Role	I will be assisting my DIPC on this project to reduce the rate of UTI associated with catheterisation. I will be joining the study team on the application, helping with review of methodology, giving advice on the measurement of catheter-associated UTI and reviewing future publications.
ORCID	0009-0002-0091-2184
% FTE Commitment	2.50%

Full Name	Dr Adrian Hopper
Organisation	NHS ENGLAND
Department	GIRFT
Proposed Role	Clinical advisor to the project (I have performed this role for previous related projects)
ORCID	0000-0002-4852-5223
% FTE Commitment	3.00%

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Full Name	Ms Patricia Mecinska
Organisation	KING'S COLLEGE HOSPITAL NHS FOUNDATION TRUST
Department	Patient and public involvement
Proposed Role	PPI co-lead
ORCID	
% FTE Commitment	5.00%

APPLICATION DETAILS

Plain English Summary

Background to the research

Catheter-associated urinary tract infection (CAUTI) is the second most common healthcare-associated infection in England, costing the NHS an estimated £604m a year (UKHSA, 2025). Risk increases every extra day a catheter stays in place, and 3-4% of cases progress to serious bloodstream infection such as sepsis (Shuman et al., 2010). These harms persist not because staff lack knowledge, but because key information – such as the insertion date – is often invisible in busy ward environments. Behavioural approaches that make information visible at the point of care have been shown to improve adherence to infection-prevention practice.

A previous Health Foundation-funded pilot at King's College Hospital tested a visible "best-before" insertion-date sticker on catheter bags. Average dwell time reduced from 13.3 to 7.4 days, rapid removals (≤4 days) increased from 22% to 37%, adherence exceeded 80%, and staff reported that the cue was simple, usable and sparked timely conversations. Cost-benefit modelling showed year-one savings with greater savings in later years. One trust with strong baseline practice showed limited change, highlighting the need to test the approach across a wider range of wards.

Research plan

This feasibility study will scale the intervention across all adult wards at King's (three sites, ~1,000 beds) covering medicine, surgery, frailty, neurosciences, trauma, liver, renal and haematology. The intervention package includes:

· Large insertion-date stickers or tube-tags

· Whiteboard reminder magnets for daily board rounds

· Posters reinforcing "No catheter, no infection"

· A 10–15-minute micro-brief

· Ward level "Catheter Champions".

Wards will adopt the intervention in phased clusters every 2–3 weeks, allowing adaptation and feedback, following a three-month baseline period. The intervention period will last six months, to explore sustained engagement. Interviews, observation and surveys with staff will capture staff acceptability, barriers and enablers to intervention adoption.

Primary feasibility outcomes include: delivery across ≥90% of eligible wards; ≥85% correct use of stickers/tags; successful linkage of dwell-time data with microbiology results; early signals of shorter dwell time and lower CAUTI rates; and staff rating the intervention as useful with ≤5 minutes added workload per shift.

Working with diverse people and communities with relevant lived experience

· The intervention has been co-designed with staff across three NHS Trusts

· A steering group with patient/carer representatives will guide the project, ensuring materials are accessible and patient experience is central.

Knowledge mobilisation

· Outputs will include an adoption paper, toolkit, and open-access resources for NHS-wide learning.

· If feasibility criteria are met, the project will progress to a substantive, multi-site study.

ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	There are no potential conflicts. The Lead Applicant is a member of the Advisory Committee on Antimicrobial Prescribing, Resistance and Healthcare-Associated Infection (APRHAI) and the Research & Development Committee of the Infection Prevention Society. The lead applicant is also a member of the consensus groups for the New Hospital Programme and the NHSE National Infection Prevention & Control Manual.

Research Plan Template for Domestic Programmes

Background and rationale

Catheter-associated urinary tract infection (CAUTI) is a predictable and preventable harm that persists despite longstanding awareness and guidance.
Each additional day a urinary catheter remains in place increases risk. A Point Prevalence Survey in acute Trusts across England found urinary tract infections to be the second most common healthcare-associated infection (HCAI), accounting for 17.5% of all cases (UKHSA, 2023). Treating UTIs costs NHS hospitals an estimated £604 million in 2023–2024, driven by antibiotic use and extended bed days (UKHSA, 2025). Around 3.6% (95% CI 3.4–3.8%) of patients with CAUTI develop life-threatening secondary infections such as bacteraemia or sepsis, with mortality rates between 10% and 33% (Shuman et al., 2010). Those most at risk include immunocompromised patients, older adults, and those with diabetes. Prolonged catheterisation also restricts mobility, particularly among frail patients, causing discomfort and loss of dignity.

Yet in busy acute settings, catheters often remain 'out of sight': insertion dates are unclear, ownership is diffuse, and prompts for review are inconsistent. Although digital systems have shifted this information online, they have not necessarily made it more visible or usable in real time. Designing interventions that are based on behaviour change theories may help to improve compliance to infection prevention & control (IPC) practice, as compliance with IPC practices is often low (Greene and Wilson, 2022). There is evidence from multiple high quality systematic reviews that interventions which include behavioural change techniques may be effective in producing modest improvements in adherence to clinical guidelines in aspects of infection control (Norman (2021). National policy emphasises regular review and timely removal, but the behaviours that sustain these practices are not always supported at the point of care (Health Innovation Network/Revealing Reality, 2020). Implementation strategies are important because reducing urinary catheter use involves changing well-established habits (Meddings et al, 2014). While longer-term technological advances - such as catheters that automatically record insertion and removal - may help eliminate this gap, immediate, low-cost and practical solutions are needed now to ensure nursing staff can easily access the information they need to do their jobs safely and effectively.

Evidence from previous work: King's College Hospital NHS Foundation Trust participated in a Health Foundation-funded behavioural science programme that co-designed and tested a low-cost intervention to make catheter duration visible and prompt earlier review. Two geriatric wards took part in the intervention, which included a 'best before' sticker applied to catheter bags highlighting the insertion date, prompting conversations about catheter removals [see accompanying document].

In two of three participating trusts, average duration of catheterisation (dwell time) fell markedly pre and post intervention (13.3 → 7.4 days at King's College Hospital Trust) and the proportion of catheters removed rapidly increased (during the intervention, 37% of catheters were removed in less than four days, compared to 22% of catheters removed in less than four days pre-intervention). Staff feedback on

sticker use was very positive - they described the large insertion-date 'best-before' sticker as a simple cue that prompted removal conversations. The adherence to sticker use was high both at Kings (81%) and the other centres in the study (78% and 96%). Cost-benefit analysis indicated year-one savings, with further savings predicted in future years once set up costs had been accounted for.

However, one of the participating trusts, the Royal Wolverhampton Hospital, showed little change, potentially reflecting a ceiling effect because due to stronger baseline practice with shorter duration of catheterisation. This underlines the importance of understanding where the intervention adds most value and how it performs in specialties with different workflows and cultures.

Building on the pilot, King's aims to assess the feasibility of rolling out the intervention hospital-wide across specialities, before progressing to a full trial.

Why is this research needed now? King's is ideally placed to assess feasibility of hospital-wide roll out. Having contributed to the original pilot, the Trust has strong internal buy-in, high ward-level engagement and new capability to link microbiology data to catheter data, producing routine ward-level CAUTI rates – something that was originally not possible during the pilot. A trust-wide feasibility study is therefore both timely and proportionate: it will test delivery across varied specialties, confirm the feasibility of using routine infection data, and identify what enables sustainable adoption at scale. This is applied, practice-driven research focused on improving efficiency and reducing barriers to information, not adding workload. The intervention integrates simple, co-designed behavioural cues into existing workflows, including handovers and board rounds, keeping catheter duration easily and efficiently 'in sight' at the point of care. It is extremely low-cost (estimated by NHS health economists at ~£176 per ward per year for the materials) and requires no additional staffing or IT build.

Trajectory to patient benefit and alignment with national priorities. Reducing catheter dwell times has the potential to reduce infections, improving patient comfort and dignity, shorten hospital stays and reduce avoidable antibiotic use – aligning directly with national priorities on infection prevention, antimicrobial stewardship and operational efficiency.

Aims and objectives

This feasibility study will test whether the package can be delivered consistently across a large acute hospital, capture robust infection outcomes through routine data collection, and define the parameters for a multi-site evaluation within 1–3 years. *Primary feasibility objectives include:*

- **Expansion across all adult wards for all specialities:** Practicality of delivering the intervention at scale across 3 hospital sites: acute specialty medicine and surgery, Frailty/Rehab, and a District General hospital site. The protocols would be adapted across different ward types, including cardiology, haematology, paediatrics, trauma network, liver, renal, neurosciences and stroke.
- **Adherence and compliance** to confirm widespread, accurate usage of materials

- **Acceptability & deliverability** from staff, and the extent to which the intervention can be embedded within existing workflows
- **Measurement of infection outcomes:** Feasibility of linking catheter dwell time data to infection data, to explore whether reduced dwell time could be linked to measurable reductions in infections

Secondary feasibility objectives include:

- **Economic feasibility:** estimate potential savings to inform business case
- **Inclusion and patient experience:** co-design patient-facing materials and explore effects on comfort, dignity and communication.
- **Blueprint for scale:** consolidate learning into a trust-wide adoption model and draft protocol for the multi-site evaluation.
- **Generate feasibility estimates** of adherence, data completeness, and outcome variability to inform sample size and power calculations for a full-scale evaluation.
- **Use findings to inform future innovations** in catheter care, including how catheter manufacturers might integrate these insights into product design.

This study is a collaboration between King's College Hospital NHS Foundation Trust (implementation site), Revealing Reality (behavioural-science partner releasing the intervention IP for NHS use), Adrian Hopper (Consultant Geriatrician and Clinical Lead for the previous research programme), Pinky Javier (Lead Nurse Urology), Patricia Mecinska (Assistant Director of Patient Experience and Public co-lead), and access to statistical support and health economists from King's College London, providing analytical oversight. Together, they ensure clinical relevance, methodological rigour, patient involvement, and a clear pathway to national rollout.

Methodology / plan

This feasibility study will take place at King's College Hospital NHS Foundation Trust, a large multi-specialty acute provider (~1,000 in-patient beds, across 3 sites). Wards will be grouped into eight to ten clusters covering acute medicine, surgery, trauma and orthopaedics, rehabilitation, paediatrics, neurosciences, cardiology, haematology, liver, renal and frailty.

Intervention specification → The intervention is a co-designed, low-cost behavioural package that makes catheter duration visible and prompts timely review and removal.
Core components, which are shown in the accompanying document, include:

1. Insertion-date sticker / tube-tag – a large, durable label on the catheter bag or tubing showing the insertion date clearly. Drawing on behavioural insight, it is designed to evoke feelings of mild disgust and urgency, and make information salient.
2. Board-round prompt – a small magnet that can be used on ward whiteboards, reminding staff to discuss catheter need during daily huddles.
3. Visual cues & posters – reinforcing the "No catheter, no infection" message.
4. Micro-brief + animation – a 10–15-minute induction covering correct dating, documentation and bag-change rules.

5. Catheter Champions – nominated housekeepers or healthcare assistants who maintain stock, monitor adherence and share feedback.

A **staggered rapid-implementation** design will test feasibility, onboarding and acceptability at scale:

Set up phase [3-4 months] Governance set up and ethics. Updating materials and protocols to maximise applicability across hospital settings.

Pre-intervention phase [5 months] Clusters will start in control for a minimum of 3 months, and cross to intervention at 2–3-week intervals until full coverage is reached. The phased approach allows manageable training, local adaptation and feedback between waves. In the participating wards, dedicated nursing staff will review each catheter daily and record the insertion date and removal date in a bespoke app created for the study. This will allow systematic monitoring of catheter dwell time pre- and post-implementation, supporting internal comparisons to evaluate feasibility and detect preliminary signals of effectiveness.

Intervention phase [6 months] The live phase will last six months, to explore sustained use of the intervention. Dedicated nurses will audit intervention compliance, recording whether a sticker is present. Clinical data sources include microbiology results (confirmed CAUTIs), urinary catheter-related blood stream infections, and EPR extractions (insertions / removals). To increase data points for less-frequent outcomes such as CAUTI, King's will extract 6–12 months of retrospective ward-level data, to strengthen trend estimation.

Process and implementation evaluation → Staff interviews, a staff survey, and observation in wards, will be conducted to assess acceptability and the practicality of delivering the intervention across wards.

Monitoring and learning → Monthly huddles with matrons, infection-prevention and data leads will monitor data completeness and emerging barriers. Rapid feedback will enable course-correction throughout and generate learning for future scale-up.

Analysis → The analysis will combine quantitative and qualitative methods to assess the feasibility, acceptability, and potential impact of implementing the intervention at scale. Quantitative analysis will use descriptive statistics to summarise adherence rates, data completeness, catheter dwell time, and infection outcomes before and after implementation, providing internal comparisons and preliminary estimates of variability to inform future sample size calculations. Qualitative data from staff and patients will be thematically analysed to explore acceptability, usability, and integration into existing workflows. Routine data linkage will be tested to assess the feasibility of capturing robust infection outcomes, while an initial cost-consequence analysis will estimate potential savings. Findings will be synthesised to produce a practical blueprint and evaluation framework for multi-site rollout.

Inclusive research: Patient and Public Involvement & Engagement

A Project Steering Group will include six patient/carer representatives from diverse backgrounds alongside clinical, behavioural-science and operational members. Meeting quarterly, it will provide independent challenge and keep delivery, ethics and communications rooted in patient experience. They will also ensure the provision of disability friendly materials such as easy read, and videos.
The intervention is visible to patients, not hidden in paperwork. Stickers and prompts act as shared cues for review timing, supporting two-way accountability and informed conversations. A co-design and challenge panel (6–8 people with lived catheter experience or caring roles) will refine patient-facing materials and shape the acceptability assessment. Sessions will be flexible (online or in person) and reimbursed under NIHR guidance. A short reflective note will record how patient input shaped design and interpretation.

Knowledge mobilisation, dissemination and impact

Dissemination will prioritise rapid NHS learning and adoption, both within King's and beyond. Outputs will include:

- An adoption paper outlining the feasibility of hospital wide intervention roll out
- A full implementation toolkit will be made available on the Trust intranet and also released openly through NIHR Open Research and the NHS England Infection Prevention and Control community of practice to guide national spread
- A short feasibility manuscript (target: *BMJ Open Quality* or *Implementation Science Communications*), a practice-oriented article for nursing and IPC networks, and short digital learning assets co-produced with staff.

Progression to a substantive study → Subject to meeting progression criteria (outlined below), the substantive multi-site study will seek funding from NIHR HS&DR (with NIHR HTA as a secondary route). The feasibility work will generate the implementation learning, effect-size estimates and adoption blueprint needed to support this application, and time is built in to draft the full protocol and outcome framework. This creates a clear pathway from feasibility → trust-wide adoption → readiness for national scale-up and definitive evaluation.

These are pragmatic stop–go indicators agreed with the Trust's executive sponsor, infection-prevention leads and analysts. Feasibility will be confirmed if:

- ≥ 90 % of eligible wards are trained and active;
- stickers or tube-tags visible on ≥ 85 % of audited catheters
- routine CAUTI data can be matched to catheter dwell time data
- trends show shorter dwell times and lower CAUTI rates
- ≥ 80 % of staff rate the intervention as useful and easy to use, with average time burden ≤ 5 minutes per shift.

References

Green & Wilson (2022) The use of behaviour change theory for infection prevention and control practices in healthcare settings: A scoping review Journal of Infection Prevention Feb 22;23(3):108–117. doi: 10.1177/17571774211066779

Meddings et al (2014) Reducing unnecessary urinary catheter use and other strategies to prevent catheter-associated urinary tract infection: an integrative review | BMJ Quality & Safety Volume 23, Issue 4

Norman, G (2021). Rapid Evidence Synthesis: Behaviour change techniques for guideline compliance in infection prevention and control. NIHR ARC Greater Manchester: University of Manchester. Microsoft Word - Behaviour change techniques in IPC_RES_new template

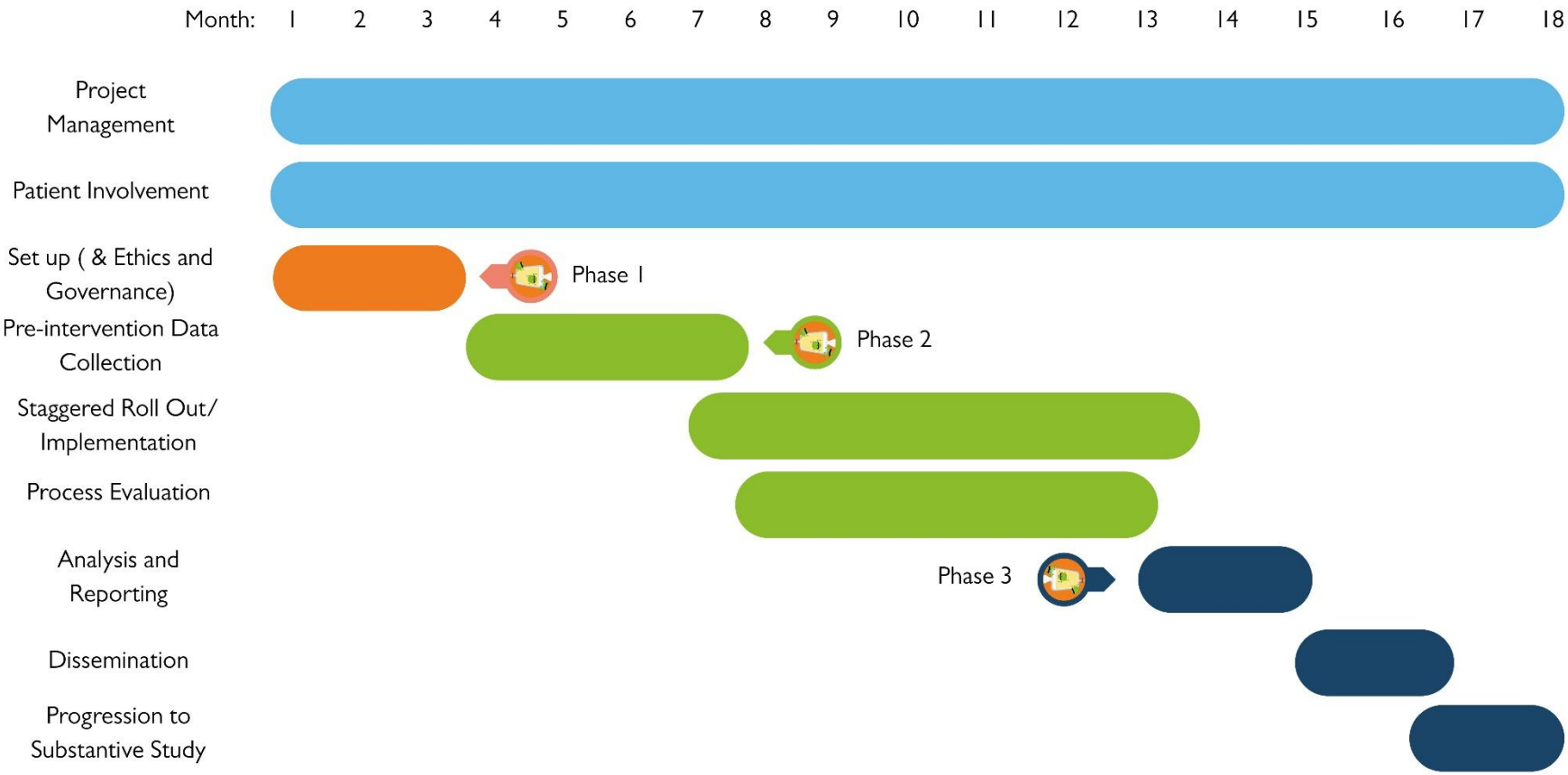
Shuman EK, Chenoweth CE. Recognition and prevention of healthcare-associated urinary tract infections in the intensive care unit. Crit Care Med. 2010;38(Suppl.): S373–S379. doi: 10.1097/CCM.0b013e3181e6ce8f. [DOI] [PubMed]

UKHSA (2023) Point prevalence survey on HCAI, AMU and AMS in England 2023: report

Point prevalence survey on HCAI, AMU and AMS in England 2023: report - GOV.UK

UKHSA (2025) Understanding the burden of UTI hospitalisations in England Understanding the burden of UTI hospitalisations in England - GOV.UK

Timings Flow Chart: Trust-wide behavioural science feasibility study on catheterisation duration and infection reduction



Intervention in Action: materials, and images from the pilot study